

Estimating Multi-label Expected Accuracy using Labelset Distributions

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Abstract—A multi-label classifier estimates the binary label state (relevant/irrelevant) for each of a set of concept labels, for a given instance. Probabilistic multi-label classifiers provide a distribution over all possible labelset combinations of such label states (the powerset of labels), from which we can provide the best estimate by selecting the labelset corresponding to the largest expected accuracy. Providing confidence for predictions is important for real-world application of multi-label models, which provides the practitioner with a sense of the correctness of the prediction. It has been thought that the probability of the chosen labelset is a good measure of the confidence of the prediction, but multi-label accuracy can be measured in many ways and so confidence should align with the expected accuracy of the evaluation method. In this article, we investigate the effectiveness of seven candidate functions for estimating multi-label expected accuracy conditioned on the labelset distribution and the evaluation method. We found most correlate to expected accuracy and have varying levels of robustness. Further, we found that the candidate functions provide high expected accuracy estimates for Hamming similarity, but a combination of the candidates provided an accurate estimate of expected accuracy for Jaccard index and Exact match.

Index Terms—Probabilistic multi-label classification, confidence, expected accuracy, entropy.

1 INTRODUCTION

MULTI-LABEL classifiers model the relationship between an observation vector and a set of label concepts. Multi-label classification is relevant to numerous domains, including categorising documents, tagging images, assigning medical diagnoses, and bioinformatics [1]. A review of some well-known multi-label methods is given in [2], and more generally a unifying view (with the general case of multi-output learning) is provided by [3].

Formally, given a set of L label concepts $\mathcal{L} = \{1, 2, \dots, L\}$, a labelset can be denoted $\mathbf{y} = [y_1, \dots, y_L] \in \{0, 1\}^L$, where $y_j = 1$ implies that the j -th label is relevant to the corresponding input vector $\mathbf{x}$ (and $y_j = 0$ implies that it is not relevant). In other words, $\mathbf{y}$ equivalently denotes a subset of label concepts. There are 2^L such possible labelsets (combinations of labels); i.e., the size of the powerset of $\mathcal{L}$ and the domain $\{0, 1\}^L$.

For any input vector $\mathbf{x}$, a *probabilistic* multi-label classifier provides a distribution $P(\mathbf{Y}|\mathbf{x})$ over all 2^L elements of the powerset. This provides a useful measure of confidence $P(\hat{\mathbf{y}}|\mathbf{x})$ for any given prediction $\hat{\mathbf{y}}$ in that set, providing an indicator of confidence in the prediction. The *accuracy* of the prediction can be measured by similarity function $S(\hat{\mathbf{y}}, \mathbf{y})$ of the prediction $\hat{\mathbf{y}}$ and the true labels $\mathbf{y}$, of which a variety has been proposed and are commonly used in the multi-label literature.

In this paper we consider: for a given posterior $P(\mathbf{Y} | \mathbf{x})$ (obtained from a probabilistic classifier, making a prediction on $\mathbf{x}$) and accuracy metric S , what is the expected accuracy? Our notation is provided in Table 1. We further clarify and motivate with an example as follows.

TABLE 1: Notation used throughout the article

Symbol	Meaning
$\mathcal{L}$	set of label concepts, $\mathcal{L} = \{1, 2, \dots, L\}$
y_j	$y_j \in \{0, 1\}$; where $y_j = 1 \Leftrightarrow$ the j -th label is relevant
$\mathbf{y}$	$\mathbf{y} \in \{0, 1\}^L$, labelset, e.g., $\mathbf{y} = [0, 1, 0, 1]$
$\mathbf{y}_i$	i -th labelset, $i = 1, \dots, 2^L$, $\mathbf{y}_i = [y_{i1}, \dots, y_{iL}]$
$\mathbf{x}^{(n)}$	n -th instance (of N in total) from the dataset
$\mathbf{y}^{(n)}$	labelset associated with n -th instance
$\hat{\mathbf{y}}$	any prediction from a multi-label classifier, $\hat{\mathbf{y}} \in \{\mathbf{y}_i\}_{i=1}^{2^L}$
$P(\mathbf{y}_i \mathbf{x})$	$\equiv P(\mathbf{Y} = \mathbf{y}_i \mathbf{X} = \mathbf{x})$; if $\mathbf{x}$ clearly implied, then $P(\mathbf{y}_i)$
$P(\mathbf{Y} \mathbf{x})$	$\equiv P(\mathbf{Y} \mathbf{X} = \mathbf{x})$; provides $(\{P(\mathbf{y}_i \mathbf{x}) \mid \forall i\})$
$\mathbf{y}^*$	MAP/mode estimate; see equation (1)
$S(\mathbf{y}_i, \mathbf{y})$	similarity function/accuracy metric; c.f. eqs. (5)–(7)
$E_S(\mathbf{y}_i)$	the expected accuracy of predicting $\mathbf{y}_i$ under metric S
$f(P(\mathbf{Y} \mathbf{x}))$	score function; confidence $\in [0, 1]$ wrt $P(\mathbf{Y} \mathbf{x})$

Example Consider the set of categories $\mathcal{L} = \{\text{Diabetes, Hypertension, SARS-CoV-2}\}$, of which a subset can be applied as a diagnosis for a particular patient described by $\mathbf{x}$. For example, if $\hat{y}_3 = 1$ there is a strong belief (diagnosis) that patient $\mathbf{x}$ has contracted the covid19 virus. Since the first two ailments are comorbidities for the third, the accurate detection of *combinations* of labels is important, implying the use of a particular accuracy metric S (such as the exact match metric, discussed below). An expression of uncertainty in the form of $P(\mathbf{Y} | \mathbf{x})$ is already provided, our objective is to estimate the expected accuracy, S of the multi-label classification $\hat{\mathbf{y}}$, that would correlate with confidence in that classification.

In other contexts, it is not necessary to consider any combination constraint and thus another metric S may be more relevant, for example Hamming Similarity (also discussed below) where true negatives are relevant, or Jaccard similarity where true negatives are not relevant. However the need for an estimate of expected accuracy remains, only for a different definition of accuracy. Indeed, in many practical use cases, any similarity metric can be used and expected

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accuracy studied, with good motivation. The expression $P(Y | x)$, made available by any probabilistic multi-label method, is what we use to estimate it. We investigate, the best way to derive that confidence; in particular, with regard to different evaluation metrics. We seek a function $f \approx E_S \in [0, 1]$.

In the literature, typically the labelset y^* associated to the mode of the multi-label distribution,

$$y^* = \arg \max_{y \in \{0,1\}^L} P(y|x) \quad (1)$$

is selected as the predicted response (including, implicitly, in non-probabilistic classifiers), and we would report $P(y^*|x)$ as the *confidence*. The remainder of the distribution is typically discarded.

Numerous methods from the literature use components of equation (1); offering some mechanism to deal with the inherent complexity. For example, the class of methods often called the ‘*label powerset* approach’ [4], [5], which implicitly replaces the powerset $\{0, 1\}^L$ with a smaller subset of much smaller dimension. The family of *classifier chains* [6] offers search mechanisms to efficiently traverse parts of the space with a greedy exploitation of the chain rule [7] or some other approximation [8]. Even when the base classifiers (responsible for individual label predictions) are not inherently probabilistic, such as a decision tree, an approximation to $P(Y|x)$ can easily be obtained via, e.g., ensemble voting.

Note that in this paper we do not focus on the issue of computational complexity, but rather take an interest in the confidence predictions themselves and more generally, the posterior joint distribution they provide: how confident can we be regarding any given prediction $\hat{y}$, and specifically the mode y^* ? In other words, given the posterior distribution $P(Y|x)$ provided by our classifier, with what confidence can we predict y^* , with regard to our chosen accuracy metric? What is the *expected accuracy* of our prediction? We answer this question in terms of the *shape* of $P(y|x)$, and we present seven candidate accuracy functions to produce an estimate. We confirm with an empirical investigation on a range of multi-label datasets and a number of different methods.

The novel contributions of this article are: • The presentation of seven candidate expected accuracy functions with suitable properties for measuring expected accuracy (Section 4). • An investigation into the accuracy of each candidate expected accuracy function for each evaluation function, with respect to data and classifier (Section 6.1). • An analysis of the robustness of the candidate expected accuracy functions with respect to data, classifier and metric (Section 6.2) • An investigation into the utility of combining expected accuracy functions (Section 6.3).

The article will proceed as follows: Section 2 gives an introduction to probabilistic multi-label learning and describes the research problem of confidence in multi-label predictions. In Section 3 we survey and discuss existing approaches in the literature and connect them to our contribution. In Section 4 we introduce the set of candidate functions that have the potential to measure expected accuracy. Sections 5–6 provide experimental results and discussion, examining the effectiveness of each candidate expected accuracy function and the effect of data and methods on each.

2 CONFIDENCE FOR PROBABILISTIC MULTI-LABEL LEARNING

In this section we provide background for discussing confidence in the context of multi-label learning and how it is a measure of expected accuracy.

2.1 Confidence in binary classification

The confidence of a probabilistic binary (single-label) classifier in its prediction $\hat{y}$ for some test instance x , is the expected accuracy for providing that prediction versus the unknown true label y . Classification accuracy is given as

$$S(\hat{y}, y) = [\hat{y} = y]$$

where $[A]$ is an indicator function, returning 1 if A is true and 0 otherwise; i.e., $S(\hat{y}, y)$ provides 1 if the prediction $\hat{y}$ is identical to the true label y . Then our confidence is

$$\begin{aligned} E_S(\hat{y}) &= \mathbb{E}[S|x] = \mathbb{E}_{Y \sim P}[[\hat{y} = Y] | x] \\ &= \sum_{y \in \{0,1\}} 1_{y=\hat{y}} P(Y = \hat{y}|x) = P(Y = y|X = x) \end{aligned} \quad (2)$$

i.e., the expected accuracy; expected value of S . Because we do not (at testing time, in practice) know the true value of y , we only have the conditional distribution $P(Y|x)$ estimated by the probabilistic binary classifier. For example, if

$$P(Y = 1|X = x) = 0.6 \quad P(Y = 0|X = x) = 0.4$$

the model is telling us to choose 1 with probability of 0.6; when randomly sampling from this distribution, there is a 60% chance that the prediction is correct and a 40% chance that it is incorrect. This label distribution portrays low confidence from the classifier. On the other hand if we have

$$P(Y = 1|X = x) = 0.9 \quad P(Y = 0|X = x) = 0.1$$

there is a 90% chance that the prediction is correct and a 10% chance that it is incorrect; this portrays high confidence in the prediction from the classifier.

High confidence is equivalent to low uncertainty. The uncertainty is expressed in the form of the expectation, which arises because we have no knowledge of the true y that random variable Y takes here.

2.2 Confidence in multi-label classification

A probabilistic multi-label classifier provides

$$P(Y|X = x) \quad (4)$$

and thus $P(Y = y_i|X = x)$ for any $y_i \in \{0, 1\}^L$, i.e., the probability of each label vector y_i to be associated to instance x . We provide an example in Table 2 (recall, also, the summary of notation in Table 1). An excellent introduction and analysis in the multi-label context is provided by [9].

In multi-label classification, there is the complication of many accuracy metrics being used in the community¹. To compute confidence for multi-label classification, we must

1. In this article we use the term ‘accuracy’ in the generic sense of comparing two labelset predictions under similarity function $S(\cdot, \cdot)$; rather than a specific sense of some multi-label papers where ‘accuracy’ is synonymous with a particular metric, often Jaccard Similarity

TABLE 2: Example of a distribution $P(\mathbf{y}|\mathbf{x})$ under an observation $\mathbf{x}$ where $L = 3$, i.e., $\{P(\mathbf{y}_i|\mathbf{x})|i = 1, \dots, 2^L\}$, showing each $\mathbf{y}_i = [y_1, y_2, y_3]$. This distribution is also shown graphically in Figure 1e. Marginal probabilities for each label, $P(Y_j|\mathbf{x})$, are shown in the final row. The expected accuracy $\mathbb{E}[\mathcal{S}]$ for the three different metrics $\mathcal{S}$ we study, are also appended alongside each labelset/prediction.

i	y_1	y_2	y_3	$P(\mathbf{y} \mathbf{x})$	$\mathbb{E}[\text{HS}]$	$\mathbb{E}[\text{EM}]$	$\mathbb{E}[\text{JS}]$
1	0	0	0	0.000	0.611	0.000	0.000
2	0	0	1	0.333	0.500	0.333	0.333
3	0	1	0	0.250	0.556	0.250	0.333
4	0	1	1	0.000	0.444	0.000	0.347
5	1	0	0	0.250	0.556	0.250	0.333
6	1	0	1	0.000	0.444	0.000	0.347
7	1	1	0	0.167	0.500	0.167	0.417
8	1	1	1	0.000	0.389	0.000	0.389
$P(y_j \mathbf{x})$	0.42	0.42	0.33				

compute the expected accuracy *with respect to the chosen metric*.

One may immediately verify that the marginal and joint modes shown in Table 2 refer to different tasks/objectives, since the marginal modes indicate a suitable prediction of $\hat{\mathbf{y}} = [0, 0, 0]$ (with a threshold of 0.5), whereas the joint mode suggests $\mathbf{y}^* = [0, 0, 1]$. Further discussion is given in, e.g., [10].

Here are the similarity metrics $\mathcal{S}$ that we consider in this paper, based on their importance and popularity in the multi-label literature (they can be found throughout many papers, e.g., [7], [4], [11] and references therein), defined over a set of N instances and L labels:

$$\text{HAMMING SIMILARITY (HS)} := \frac{1}{NL} \sum_{n=1}^N \sum_{j=1}^L [y_j^{(n)} = \hat{y}_j^{(n)}], \quad (5)$$

$$\text{EXACT MATCH (EM)} := \frac{1}{N} \sum_{n=1}^N [\mathbf{y}^{(n)} = \hat{\mathbf{y}}^{(n)}], \text{ and} \quad (6)$$

$$\text{JACCARD SIMILARITY (JS)} := \frac{1}{N} \sum_{i=1}^N \frac{|\mathbf{y}^{(n)} \wedge \hat{\mathbf{y}}^{(n)}|}{|\mathbf{y}^{(n)} \vee \hat{\mathbf{y}}^{(n)}|}. \quad (7)$$

where $\mathbf{y}^{(n)}$ is the labelset associated to the n th instance $\mathbf{x}^{(n)}$, where $\wedge$ and $\vee$ are the bitwise logical AND and OR operations, respectively. Other notation is clarified in Table 1.

The expected multi-label accuracy, extending Eq. (3), is

$$\begin{aligned} E_{\mathcal{S}}(\mathbf{y}^{(n)}) &= \mathbb{E}[\mathcal{S}|\mathbf{x}^{(n)}] = \mathbb{E}_{\mathbf{Y} \sim P(\mathbf{Y}|\mathbf{x})}[\mathcal{S}(\mathbf{y}^{(n)}, \mathbf{Y})|\mathbf{x}^{(n)}] \\ &= \sum_{\mathbf{y}_j \in \{0,1\}^L} P(\mathbf{y}_j|\mathbf{x}^{(n)}) \mathcal{S}(\mathbf{y}^{(n)}, \mathbf{y}_j) \end{aligned} \quad (8)$$

which produces the final columns in Table 2, according to each of the metrics of our study. The expected accuracy with respect to evaluation function $\mathcal{S}$ can be expressed as “If I randomly choose a labelset according to the labelset distribution, what accuracy do I expect when evaluating using $\mathcal{S}$.” For example, consider that the true labelset in Table 2 is $[1 \ 0 \ 0]$, then the expected Jaccard accuracy is 0.333. Note that if the labelset distribution contains high certainty and is correct, the expected accuracy will be greater than a distribution with low certainty. To illustrate with

an example: Figure 1 shows six hypothetical distributions $P(\mathbf{Y}|\mathbf{x})$ over the powerset of three labels, conditioned on some hypothetical instances $\mathbf{x}$. For the prediction, we could provide $\mathbf{y}^*$, the mode labelset of $P(\mathbf{Y}|\mathbf{x})$ known to maximise exact match similarity. For confidence in this prediction we could return $P(\mathbf{y}^*|\mathbf{x})$. Intuitively we would expect that distribution (a) results in a low expected accuracy since all labelset combinations are equally unlikely, whereas distribution (f) should result in high expected accuracy; yet it is not obvious how we should measure the expected accuracy from the remaining distributions. The goal of this paper is to formalise and test such intuitions, and provide a function f with which to do so.

But, in practice, the true labelset is not known (it is the task of the classifier to predict the true labelset), so we are not able to directly compute the expected accuracy.

In this paper, we examine if we are able to estimate the expected accuracy (confidence with respect to the evaluation function) based on the labelset distribution. More specifically, we investigate if there is a relationship between the set of candidate functions f applied to the labelset distribution and expected accuracy, for different evaluation functions and data.

3 RELATED WORK

In this section we look at related methods for making decisions under uncertainty in context of multi-label classification, as relating to a joint-posterior distribution, Eq. (4).

3.1 Thresholding

A simple and common approach to gain trust in a probabilistic method is to calibrate a threshold according to desired confidence [12]; i.e., we can accept $\mathbf{y}_i$ if $P(\mathbf{Y} = \mathbf{y} | \mathbf{X} = \mathbf{x}) > t$ where t some ad-hoc threshold. However, unlike in the simple binary case, there is no obvious value such as 0.5. Indeed, the probability is probably quite low for many $\mathbf{y}_i$, as in the examples in Figure 1.

3.2 Ranking

Rather than attempting an explicit separation between relevant and irrelevant labels, authors have instead looked at *ranking* labels [13], [14]. An intrinsic label-wise ranking is readily available via their marginal scores $P(y_j|\mathbf{x})$ (recall the example from Table 2, which corresponds to two equivalent best rankings: $(1, 2, 3) \equiv (2, 1, 3)$ from best to worst, wrt label index).

3.3 Reject and abstention

Classification with a *reject option* (also known as abstention), has been an active research area [15], [16], [17]. This can be seen as an extension of thresholding, where low confidence leads to abstention (no decision). A new and interesting look at abstention in the context of multi-label classification is given in [18], building on [19] where the rejection region was defined for each label using an upper and lower bound, and computed using a cost function based on a predefined cost for rejection. In the multi-label setting one can consider partial abstention [20], where a model may deliver predictions for a subset of class labels.

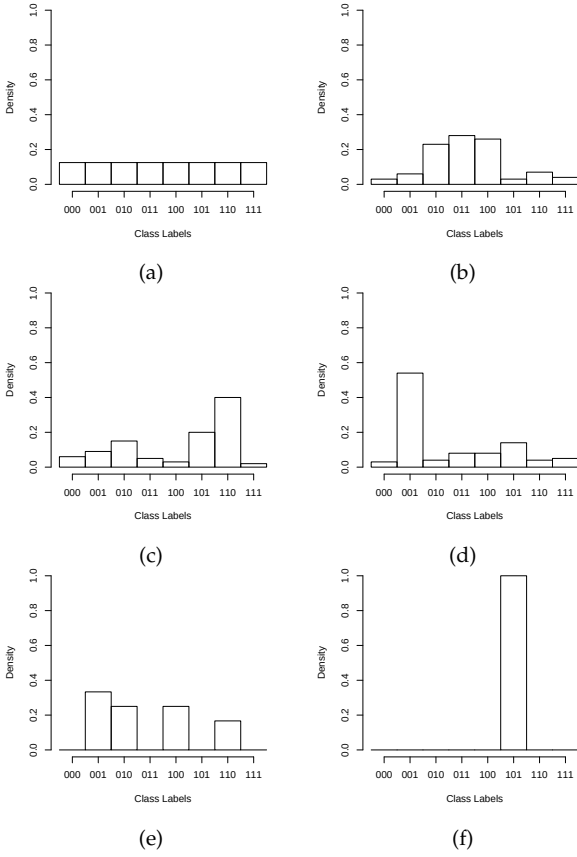


Fig. 1: Six possible multi-label distributions $P(\mathbf{y}|\mathbf{x})$ over three labels for a hypothetical given $\mathbf{x}$. It is clear that the top left distribution should provide the lowest expected accuracy (and correspondingly, the greatest uncertainty), and the bottom right distribution should provide high expected accuracy and the least uncertainty regarding classification, particularly with regard to exact match similarity, but it is not clear what level of expected accuracy we should provide to the remaining distributions. Distribution (e) replicates graphically the distribution given in Table 2.

In this work, we do not target this specific reject option, but rather the expression of confidence associated to a given labelset prediction.

3.4 Cautious and conformal prediction

One may consider being *cautious* (or, equivalently, *imprecise*) about predictions when uncertainty is high, and provide a set of possible labellings in the form of multiple predictions or posterior distributions $\{P(\mathbf{Y}|\mathbf{x})^m\}_{m=1}^M$ rather than a single one (the index m is for an entire distribution); and rather than simply accepting the most likely labelset, or refusing to make a decision. In other words, the uncertainty for a given prediction is described by this *credal set*. These are also known also as *distributionally robust* models, and they have recently been investigated in the multi-label case [21], [22].

Beyond representing uncertainty with a set of probability distributions, *conformal prediction* will use the empirical distribution to determine the confidence levels of prediction

sets [23]. This brings us closer to the question we tackle in this study: how to gauge the *potential accuracy* of a given labelset prediction/combination $\mathbf{y}$ in reference to a given distribution/model $P(\mathbf{Y}|\mathbf{x})$. In other words, we study the calibration of expected accuracy to confidence.

3.5 Other connections

In the literature of particle filters (i.e., sequential important sampling; recursive Bayesian estimation; see, e.g., [24]) in the signal processing community one finds the need to measure the *degeneracy* of a distribution. A degenerate distribution is one which is flat, e.g., Figure 1a. A typical choice in particle filters is the effective sample size

$$\text{ESS} = \left[\sum_{i=1}^n P(\mathbf{y}_i)^2 \right]^{-1}$$

(we have adapted the notation to our multi-label prediction setting) which is along the vane of what we study in the following section (indeed, ESS is essentially a form of collision entropy without the log); as we investigate the utility of the using the multi-label distribution for each instance to estimate the expected accuracy of the predicted labelset, where expected accuracy is measured with respect to the chosen similarity function. An analogy between multi-label prediction and particle filters was noticed in [25], but did not explore variations of degeneracy metrics.

4 EXPECTED ACCURACY CANDIDATES FOR MULTI-LABEL CLASSIFICATION

We aim to use the posterior distribution $P(\mathbf{Y}|\mathbf{x})$ to provide a score representative of expected accuracy,

$$f(P(\mathbf{Y}|\mathbf{x})) \in [0, 1] \quad (9)$$

or more specifically $f(\mathbf{x})$, where a value close to 1 indicates that we expect to achieve high accuracy by predicting $\hat{\mathbf{y}}$ for $\mathbf{x}$ (using $P(\mathbf{Y}|\mathbf{x})$ as per equation (8)) and a value close to 0 indicates that this prediction $\hat{\mathbf{y}}$ is almost certainly wrong.

In other words, we hypothesise that the expected accuracy is a function of the multi-label label powerset distribution, but it is not clear how to map the distribution to an expected accuracy value (i.e., what is the appropriate f).

We approach this problem of determining f using traditional scientific method, where we propose a set of functions $\{f_s\}$ corresponding to statistics of the distribution, and examine if the statistics provided by the candidate functions are associated to expected accuracy, when taking into account the environmental conditions (such as the data, and parameters used). If an association is found, it provides evidence that f_s is a component of f .

In Section 4.1, we will examine seven functions of this categorical distribution that may be useful for estimating the expected accuracy, in equation (9). Then in Section 4.2, we examine the bias in sample entropy and discuss why we refrain from sampling in these experiments.

4.1 Candidate Expected Accuracy Functions

We desire an *expected accuracy* function f that, given a labelset distribution $P(Y|x)$, returns a measurement of the distribution indicating the expected accuracy of the prediction for a given instance. There are two conditions that the function must satisfy:

- 1) $f = 0$ if all labelsets have equal probability (e.g., in Figure 1a), implying no certainty.
- 2) $f = 1$ if the distribution contains one labelset with probability 1 and all others probability 0 (e.g., in Figure 1f), implying total certainty.

It is not clear what f should provide when presented with other distributions, but we believe the function should be related to the certainty of the distribution.

In the following section, we examine candidate accuracy functions f that meet these constraints and also provide suitable certainty values $f \in (0, 1)$ for “in-between” distributions (such as the four other distributions in Figure 1). In our experiments, we will examine these functions in detail to determine which provide values that are most closely associated to the expected accuracy for multi-label prediction using a given similarity metric. Note that here we present the normalised form of each function. Details of the normalisation are provided in Appendix A.

In terms of computational complexity, each of the candidate functions presented below are order $O(2^L)$, since they require one scan through the labelset distribution.

4.1.1 High probability (HP)

The de facto standard for measuring confidence of multi-label predictions due to its use in other forms of classification and minimisation of the Exact Match metric. It is simply the probability of the labelset that has the greatest probability (mode).

$$C_{HP} = \max_{\mathbf{y}_i \in \{0,1\}^L} \frac{2^L P(\mathbf{y}_i) - 1}{2^L - 1} \quad (10)$$

where L is the number of labels. For example, for 3 labels, a uniform distribution of $P(\mathbf{y}_i) = \frac{1}{8}$ (since there are 8 possible combinations) gives $C_{HP} = \frac{2^3 \cdot \frac{1}{8} - 1}{2^3 - 1} = 0$; the lowest expected accuracy. Since this function is the most commonly used, it is used as the baseline in each experiment.

4.1.2 Top Gap (TG)

HP only examines the mode of the multi-label distribution. However, it may well be that the probability for other labelsets are very close to this maximum value, in which case we can be less confident about a correct prediction, as exemplified in Figure 1b.

We also consider as a candidate the difference in probability of the most probable label combination and the second most probable label combination. We call this function *Top Gap* (TG), which is given as:

$$C_{TG} = \left[\max_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i) \right] - \left[\max_{\mathbf{y}_i \in \{\{0,1\}^L \setminus \mathbf{y}^*\}} P(\mathbf{y}_i) \right] \quad (11)$$

The intuition behind the function is that a labelset distribution with a large Top Gap implies greater confidence and

hence higher expected accuracy, since one labelset has a much greater probability than the others.

If C_{TG} is small, then there is little difference between the top two probabilities, meaning that there is uncertainty as to which of the associated label combinations is the best choice. If C_{TG} is large, then there is a large difference in probability between the most probable label combination and all of the other label combinations, meaning that there is a clear choice from the probability distribution.

4.1.3 Shannon Entropy (SE)

Shannon entropy [26] is a measure of the information content from a channel, given the distribution of symbols transmitted through the channel. The greater the uncertainty of the symbols, the greater the information content.

To obtain an expected accuracy candidate [27], we subtract the normalised entropy score from 1 to obtain

$$C_{SE} = 1 + \sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i) \log_{2^L} (P(\mathbf{y}_i)) \quad (12)$$

where $C_{SE} \in [0, 1]$ is a measure of expected accuracy using Shannon Entropy ($C_{SE} = 0$ implies no confidence and low expected accuracy, while $C_{SE} = 1$ implies full confidence and high expected accuracy for the prediction).

4.1.4 Collision Entropy (CE)

Collision entropy (CE) is a commonly used form of Rényi entropy, with parameter $\alpha = 2$. The collision entropy based accuracy function is:

$$C_{CE} = 1 + \log_{2^L} \sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i)^2 \quad (13)$$

4.1.5 Min Entropy (ME)

The amount of information in (number of bits required to represent) an object is measured as the log of the reciprocal of the object's probability. Shannon Entropy is the expected value of this information measure across all possible objects. Rather than measuring the expected value (mean), we can measure the minimum information across all objects using Min Entropy (ME). The min entropy based acceptance function is:

$$C_{ME} = 1 + \log_{2^L} \max_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i) \quad (14)$$

where $C_{ME} \in [0, 1]$, $C_{ME} = 0$ implies low expected accuracy, while $C_{ME} = 1$ implies high expected accuracy (full confidence in the prediction).

4.1.6 Gini Impurity

Gini impurity (GI) is a common measure of impurity, used to determine the branching variables in a decision tree. Gini impurity increases as the distribution approaches Uniform and provides a score of zero for distributions containing probabilities of zero for all but one item. Gini impurity based accuracy function is:

$$C_{GI} = 1 - \sum_{\mathbf{y}_i \in \{0,1\}^L} \frac{P(\mathbf{y}_i)(1 - P(\mathbf{y}_i))}{1 - 2^{-L}} \quad (15)$$

where $C_{GI} = 0$ when all labels have the same probability, and $C_{GI} = 1$ when all but one item have zero probability.

4.1.7 Chi-squared Statistic

The Chi-squared statistic (CS) is a measure of difference between the set of expected and observed sample frequencies from a multinomial distribution. The Chi-squared accuracy function is:

$$C_{CS} = \sum_{\mathbf{y}_i \in \{0,1\}^L} \frac{(P(\mathbf{y}_i) - 2^{-L})^2}{1 - 2^{-L}} \quad (16)$$

A comparison of the behaviour of the seven candidate expected accuracy functions C_{HP} , C_{TG} , C_{SE} , C_{CE} , C_{ME} , C_{GI} and C_{CS} , on a multinomial distribution of size three, is shown in Figure 2. The left figure shows the effect of changing the first and second probabilities while holding the third to 0. The normalisation has forced each of the functions to have a score of 1 when the first or second probability is 1 (meaning the remaining probabilities are 0). For the middle plot, the third probability is $1/3$. We see that all functions have a value of 0 when all probabilities are equal (all $1/3$). The plot on the right has the third probability set to $1/2$. The three plots show that High Prob and Top Gap are linear except at the cusp points, Min Entropy is concave, while the remaining functions are convex. All functions show the same decrease then increase shape, except for Top Gap in the right plot, which increases then decreases (due to it only observing the difference between the top two probabilities). We also find that High Prob and Min Entropy are flat in the right plot, since they only depend on the greatest probability, which does not change for this plot. Finally, we find that Chi-Squared and Gini Impurity have provided equal expected accuracy for all probability values in all plots. In fact, we find that Chi-Squared and Gini Impurity functions are equivalent, and so only one will be used in our analysis. Details of the equivalence is provided in Appendix D.

4.2 Bias in sample entropy

Estimating the expected accuracy of the multi-label distribution using the seven candidates requires us to first compute the probability of each labelset combination in the 2^L size categorical distribution. Computing this distribution can be time consuming, depending on the number of labels. For example, given 50 labels, we have a categorical distribution with $2^{50} = 1.1259 \times 10^{15}$ items. If it takes a millisecond to compute the probability of one item, it will take 35702 years to compute the set of 2^{50} probabilities. To avoid this problem, we can either limit ourselves to data with a smaller number of labels, or approximate the distribution.

Each multi-label distribution can be approximated by sampling from the conditional distribution of each label, to obtain a sample from the multi-label distribution. But unfortunately, sample estimates of entropy are biased. Analysis has shown how to correct for such bias [28], [29], [30] by approximating the label distribution with a Poisson distribution, but this approximation is not useful for categorical distributions with a huge number of categories relative to the sample size. It is easy to see that the problem comes from estimating the long tail probabilities. For example, using 50 labels, leading to a categorical distribution containing 1.1259×10^{15} items, a sample of size 1,000,000 will miss

many of the items in the tail and underestimate their contribution to be 0 ($0 \times \log 0$). A simulation shows that the bias correction provides a slight improvement for the smaller data sets (If the true entropy is 3.960, the sample entropy is 3.901 and the bias correction gives 3.956), but for the larger data sets it should not be used (If the true entropy is 15.056, the sample entropy is 8.236 and the bias correction gives 562.911). Further research must be performed to identify how we can avoid or correct for the bias in sample entropy, but we will leave that for future work.

During our analysis, we want to control as many of the variables as possible to ensure that our experimental results are meaningful and not confounded with unknown effects. We do not want this sample bias to be a factor that influences this analysis, therefore, we will avoid sampling and compute the whole distribution, but limit the data to manageable sizes. Table 3 shows information about each of the data sets used; all labels were used for all data sets except for the Enron data, where the first 20 labels were used.

5 EXPERIMENTAL SETUP

In this section we describe the experimental setup. Then, in Section 6 we describe and discuss the experimental results with regard to absolute expected accuracy.

An extensive experimental evaluation is conducted to investigate which candidate functions (of those detailed in Section 4.1) are associated to the expected accuracy, and in which context (which kind of dataset, for what kind of classification method, and under which evaluation metric).

We use three probabilistic methods involving independence (BR), some dependence (CT) and full-dependence (ECC) modelling, described and referenced in Section 1; thus encompassing the full range of possible dependence modelling of the labels. We use three evaluation metrics – Exact Match, Hamming Similarity and Jaccard Similarity – as described in Section 2.2.

Datasets are listed in Table 3; available from the MULAN², MEKA³, and STARE⁴ repositories. Note again that we have selected data containing no more than 25 labels to avoid sampling bias (described in Section 4.2). This is not an article about extreme multi-label classification (for reference, see, e.g., [31], therefore we do not consider scalability a primary concern.

Thus, the variables involved in our experiment are: 1) the candidate expected accuracy function, 2) the data set, 3) the accuracy metric; and 4) the method of multi-label classification.

The experiments investigate the effect of each variable on the correlation between the set of scores provided by each candidate function and the true evaluation score with respect to the chosen evaluation function. When the correlation is measured using Kendall’s τ , we are measuring *relative accuracy*, when the correlation is measured using Pearson correlation, we are measuring *absolute accuracy*. Relative accuracy is a weaker association than absolute accuracy since it only measures if the estimated and actual scores

2. <http://mulan.sourceforge.net/datasets-mlc.html>

3. <https://sourceforge.net/projects/meka/files/Datasets/>

4. <http://www.ces.clemson.edu/~ahoover/stare/>

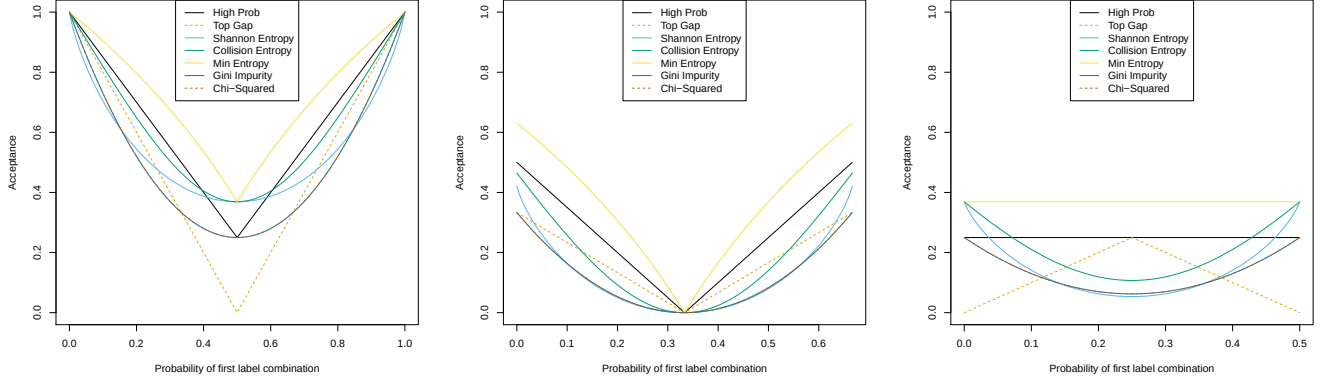


Fig. 2: A comparison of the seven candidate accuracy functions over a probability distribution with three elements. The x axis shows the probability associated to the first of the three elements, the probability of the third element is 0 for the left plot, $1/3$ for the middle plot and $1/2$ for the right plot; the y axis shows the expected accuracy provided by the candidates. All functions are normalised to provide a value of 1 when all but one item has probability zero, and a value of zero when all items have equal probability.

TABLE 3: Data sets and associated statistics (N is number of observations, L is number of labels, M number of features, LC label cardinality). Note that Enron provides 53 labels but only 20 were used in these experiments.

	N	L	M	LC	Type
Emotion	593	6	72	1.87	audio
Scene	2407	6	294	1.07	image
Yeast	2417	14	103	4.24	biology
Stare	373	15	44	1.32	medical
Enron	1702	20 (53)	1001	3.38	text
Slashdot	3782	22	1079	1.18	text

are in a similar order, whereas absolute accuracy measures if the score are linearly related. The preliminary investigation in to Relative accuracy is provided in Appendix F. The investigation in to Absolute accuracy is provided below.

6 ABSOLUTE EXPECTED ACCURACY

We now examine the potential for each candidate expected accuracy function to provide *absolute* accuracy, where the value of the candidate expected accuracy is an estimation of the accuracy of the prediction. An accurate absolute accuracy score provides us with a good estimate of the accuracy of a prediction. To measure the accuracy of each candidate for absolute accuracy, we measure the Pearson correlation between the candidate expected accuracy estimate and the accuracy of the prediction. A high correlation implies that there is a linear relationship between the prediction accuracy and the candidate expected accuracy.

6.1 Analysis 1: Marginal Correlation Analysis for Absolute Accuracy

We begin our evaluation of each candidate expected accuracy function with a marginal analysis of each of the experiment variables. We examine the response of the Pearson correlation between prediction accuracy and acceptance function. A high Pearson correlation implies that there is

a linear relationship between the candidate score and the accuracy of a given prediction, which in turn tells us that we are able to use the candidate to estimate the expected accuracy of the prediction. Therefore, we want to find which of the candidates leads to the greatest Pearson correlation.

To examine the effects on correlation, we use a fixed effects model for each evaluation function, with correlation as the response, and candidate function, data, and classifier types as the independent variables. Observing the significance of the model coefficients will identify relationships to the correlation. Fisher’s Z transform [32] was used to stabilise the variance of the correlation, and so we examined the effect of each acceptance parameter, data set, and classifier on the transformed correlation. Results are presented in Table 4.

The design matrix was constructed to ensure that all coefficient estimates are relative to HP, the Emotions data and the ECC method. Each Expected Accuracy Bias, Data Bias and Classifier Bias is the change in mean Fisher transformed Pearson correlation between the predicted expected accuracy and true expected accuracy; a positive bias means that the change of variable from the baseline has lead to an increase in correlation, meaning that the candidate accuracy function is able to provide a better estimate of expected accuracy.

The *Expected Accuracy* portion of the table shows that TG (Top Gap) is highly significant for Exact, Hamming and Jaccard Similarity and the coefficients are negative; thus, Top Gap is significantly worse than High Prob (HP). The only other statistically significant coefficient is for SE when using Jaccard Similarity (positive).

In the *Data* section, we find that Scene, Yeast and Enron all have a significant effect when using each of the Exact, Hamming and Jaccard similarities, and that Stare and Slashdot also have a significant effect when using Jaccard Similarity.

In the *Classifier* portion of the table, all of the coefficients are negative, implying that changing from ECC to either Independent or CT will reduce the correlation between

TABLE 4: The expected change in variance stabilised Pearson’s correlation for the three variables (Expected Accuracy function, Data, Classifier) and standard error in parentheses, with respect to each variable’s baseline, independent of the other variable values. A value of 0 represents no change in the correlation when changing from the baseline to the chosen item, a positive value shows an increase in correlation, a negative value implies a decrease in correlation. The stars represent the statistical significance level of each expected change. The footer of the table shows the goodness of fit statistics for the analytical model.

	Exact	Jaccard	Hamming
<i>Expected Accuracy Function Bias (Baseline: High Prob)</i>			
TG	−0.073 (0.026)***	−0.093 (0.023)***	−0.096 (0.017)***
SE	0.000 (0.026)	0.067 (0.023)***	0.025 (0.017)
CE	−0.006 (0.026)	0.029 (0.023)	0.024 (0.017)
ME	−0.024 (0.026)	−0.017 (0.023)	−0.010 (0.017)
CS	0.011 (0.026)	0.037 (0.023)	0.024 (0.017)
<i>Data Bias (Baseline: Emotions)</i>			
Scene	0.117 (0.026)***	0.054 (0.023)**	0.113 (0.017)***
Yeast	0.214 (0.026)***	0.102 (0.023)***	0.126 (0.017)***
Stare	−0.028 (0.026)	−0.043 (0.023)*	−0.004 (0.017)
Enron	0.073 (0.026)***	−0.464 (0.023)***	0.136 (0.017)***
Slashdot	−0.032 (0.026)	−0.113 (0.023)***	−0.020 (0.017)
<i>Classifier Bias (Baseline: ECC)</i>			
Independent	−0.071 (0.019)***	−0.089 (0.016)***	−0.087 (0.012)***
CT	−0.019 (0.019)	−0.025 (0.016)	−0.056 (0.012)***
R ²	0.634	0.900	0.771
RMSE	0.079	0.070	0.051

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

accuracy and acceptance, where Independent leads to a greater statistically significant drop.

Finally the goodness of fit statistics ($R^2 = 1 - SSR/SST$ measuring the proportion of the response variance captured by the model⁵ and $RMSE = \sqrt{\sum_i \epsilon_i^2}$ measuring the magnitude of the residuals ϵ_i ⁶), show that the three variables (Expected Accuracy function, Data and Classifier) explain the variability in correlation very well when using Jaccard Similarity (shown by the high R^2 value), followed by Hamming and Exact. The R^2 value for Exact is on the low side, indicating that there may be additional variables or interactions that we have not taken into account. The small RMSE values give an indication of the expected analytical model error, where all are similar. This leads us to believe that the low R^2 value for Exact is due to the accuracy-candidate expected accuracy correlation having greater total variance.

Thus, results indicate that SE should be used when measuring acceptance for Jaccard Similarity. There was no significant evidence for using other acceptance parameters over the baseline HP.

6.2 Analysis 2: Robustness Analysis for Absolute Accuracy

Robustness is very important when measuring expected accuracy, since we require that the results have only small variation based on minor changes in experimental parameters (i.e., the correlation between the accuracy and candidate expected accuracy should not be effected by a change in data, method, or evaluation function, etc). In other words, we want these different parameters to have negligible effect

on the correlation between the classifier accuracy and the estimated expected accuracy.

For this analysis, we examined each candidate function independently, and examined the effect of several data features, change in classifier, and change in evaluation function on the candidate expected accuracy to accuracy. The results of the analysis are shown in Table 5;

The section *Data Features Expected Gradient* within Table 5 contains Label Count (the number of labels in the multi-label data), Label Comb (the number of unique label combinations found in the data), Label Card (the average label cardinality for the data set), and Feature Count (the number of prediction features in the data). Each of these are either positive real or positive integer variables, therefore the estimated analytical model coefficient for each is its gradient (the expected increase in Fisher transformed correlation when the associated variable increases by 1). For example, note that the coefficient for Label Count in the HP column is −0.020, telling us that we expect the Fisher transformed correlation for HP to decrease by 0.020 with each additional label in a dataset, implying that there is a dependence of the correlation on the the number of labels in the data.

Note that we are not concerned about the baseline being non-zero; to show robustness, we are required to examine if changes to the experiment variables changes the correlation therefore we are only concerned with the coefficients of the remaining additive effects. An ideal expected accuracy function will have zero for all coefficients (implying that any change in experiment variables does not effect its correlation to accuracy).

TG is affected by most of the variables (being significant at the 5% level for most coefficients). Change in any of the variable values for both CE and CS has no significant effect

5. https://en.wikipedia.org/wiki/Coefficient_of_determination

6. https://en.wikipedia.org/wiki/Root_mean_square_deviation

TABLE 5: The expected change in variance stabilised Pearson’s correlation when adjusting four data features (Label Count, Label Comb, Label Card, Feature Count), or changing Classifier or Similarity Function (standard error in parentheses), independent of the other variable values. A value of 0 represents no change in the correlation when changing the Data Feature, or when changing from the baseline to the chosen item, a positive value shows in increase in correlation, a negative value implies a decrease in correlation. The stars represent the statistical significance level of each expected change. The footer of the table shows the goodness of fit statistics for the analytical model.

	HP	TG	SE	CE	ME	CS
<i>Data Features Expected Gradient (Zero gradient means no effect)</i>						
Label Count	-0.020 (0.031)	-0.079(0.030)**	0.029 (0.023)	0.009 (0.029)	-0.039 (0.030)	0.028 (0.029)
Label Comb	0.003 (0.009)	0.021(0.009)**	-0.010 (0.007)	-0.006 (0.008)	0.008 (0.008)	-0.011 (0.008)
Label Card	-0.068 (0.388)	-0.902(0.381)**	0.506(0.295)*	0.302 (0.364)	-0.295 (0.376)	0.525 (0.365)
Feature Count	-0.000 (0.001)	-0.002(0.001)**	0.001(0.001)*	0.001 (0.001)	-0.001 (0.001)	0.001 (0.001)
<i>Classifier Bias (Baseline: ECC)</i>						
Independent	-0.100(0.045)**	-0.134(0.044)***	-0.032 (0.034)	-0.068 (0.042)	-0.104(0.043)**	-0.055 (0.042)
CT	-0.046 (0.045)	-0.090(0.044)**	0.026 (0.034)	-0.026 (0.042)	-0.053 (0.043)	-0.010 (0.042)
<i>Similarity Function Bias (Baseline: Exact)</i>						
Hamming	-0.025 (0.045)	-0.048 (0.044)	-0.000 (0.034)	0.005 (0.042)	-0.010 (0.043)	-0.011 (0.042)
Jaccard	-0.078(0.045)*	-0.098(0.044)**	-0.012 (0.034)	-0.044 (0.042)	-0.072 (0.043)	-0.052 (0.042)
R ²	0.424	0.459	0.261	0.341	0.453	0.327
RMSE	0.134	0.131	0.102	0.126	0.130	0.126

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

on the accuracy-confidence correlation. Both HP and ME show a significant effect for changing the classifier from ECC to Independent and HP has the addition significant effect when changing from Exact to Jaccard Similarity. SE shows to have a significant effect on two of the data variables (Label card and Feature count), but it is interesting to see that this effect is positive, meaning that the correlation between the candidate function and accuracy when using SE increases as the Label cardinality or Feature count increases.

Examining the Classifier variable, we find that these results reinforce our belief that the accuracy-candidate correlation is dependent on the complexity of the classifier.

Surprisingly, SE has the lowest R^2 . therefore the accumulation of all variables must explain more of the variance for CE and CS when compared to SE, even though SE has two significant coefficients.

Hence we see that CE and CS are the most robust in this context. SE was affected by some of the data parameters, but it showed that the correlation improves as the Label cardinality increases, which is usually a function of data size. Therefore, SE should be better suited to large data sets than small data sets.

6.3 Analysis 3: Mixture Analysis for Absolute Accuracy

So far we have examined the potential for each of the candidate expected accuracy function to be used to measure absolute acceptance (we examined correlation, showing a linear relationship), but we have not examined how to map the candidate functions to an estimated accuracy. In this section, we will fit models to estimate the Exact, Hamming and Jaccard accuracy using the candidate functions, to assess how well each is able to provide estimates of absolute accuracy.

To estimate the expected accuracy, we build a logistic regression model. The features of the model are the predicted expected accuracy provided by the candidate function (continuous), the data set name (categorical), and

the classification method (categorical). Since the accuracy results vary heavily based on the data set, we also included an interaction term between the candidate function and data set, giving the expected accuracy model:

$$\text{logit}(p_{ijx}) = \beta_i \text{candidate}(x) + \text{data}_i + \text{classifier}_j \quad (17)$$

where $\text{candidate}(x)$ is the value provided by the candidate function for instance x , β_i is the candidate coefficient for data set i , data_i is a bias term for data set i (i is either Emotions, Scene, Yeast, Stare, Enron, or Slashdot), classifier_j is the bias term for classifier j (j is either Independent, CT or ECC), logit is the logit function, and p_{ijx} is the estimated accuracy of the prediction for instance x .

Both Hamming Similarity and Exact Match are suited to logistic regression, since they are both proportions (Exact is either 1 or 0, and Hamming is represented as the proportion of correct labels). Modelling Jaccard Similarity is not as straightforward. Since both the numerator and denominator (see equation (7)) are affected by the prediction, we can not simply model the similarity as a proportion. We can instead decompose the Jaccard Similarity into the number of true and false positives and negatives:

$$S_{\text{Jaccard}}(y, \hat{y}) = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \quad (18)$$

where TP is the number of labels that were predicted as positive and are actually positive, FP is the number of labels that were predicted as positive, but are actually negative, FN is the number of labels that are predicted as negative, but are actually positive, thus $\text{TN} = N - \text{TP} - \text{FN} - \text{FP}$ is the number of labels that were predicted as negative and are actually negative (N is the number of labels). The number of labels N is constant for each prediction, therefore the set $\{\text{TP}, \text{TN}, \text{FP}, \text{FN}\}$ is equivalent to a multinomial distribution with N trials. Modelling this multinomial distribution will provide estimates for the four true/false positive/negative

rates p_{TP} , p_{TN} , p_{FP} , p_{FN} allowing us to model the Jaccard Similarity finally as

$$S_{\text{Jaccard}}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{p_{TP}}{p_{TP} + p_{FP} + p_{FN}} \quad (19)$$

using a multinomial logistic regression with the same covariates as in equation 17. The candidate function scores can then be used to predict the true/false positive/negative proportions and then reconstruct the estimate of the Jaccard Similarity.

Using the predictive models for Exact, Hamming and Jaccard Similarity, we examine the utility of HP, SE and CE candidate functions (which so far showed merit) to predict the accuracy. We will also examine the accuracy of a combined model using the model in equation 17 where all candidate functions are linearly combined, rather than only one.

The complete data set (containing all multi-label data sets) was randomly split into 50% training and 50% testing sets and 10-fold cross validation was used on the training set to determine the ridge regularisation parameters of each model. The complete randomised training/testing process was replicated 20 times to obtain distributions of expected accuracy for each fitted model.

To evaluate the results, we must remember that each predicted value is an expected accuracy value; we can evaluate the prediction by comparing it to the actual accuracy for the given instance. For our evaluation, we are also interested in the value of the actual accuracy (e.g. there will be cases when we want a more accurate model for only the high accuracy results rather than all results). So we provide measurements for various ranges of accuracy. The estimated expected accuracies were divided into 0.1 intervals and the associated instances in each interval were used to compute the true expected accuracy (the intervals were size 0.05 for Hamming confidence due to the smaller range of scores). This was applied to the 20 replicates to obtain a distribution over the expected accuracy. The 95% interval of the expected accuracy distribution is reported for each of the Hamming, Jaccard and Exact Match confidence score intervals in Tables 6, 7 and 8 respectively. Ideally, the 95% expected accuracy interval should be contained within the given estimated expected accuracy interval.

First examining the Hamming table (Table 6), the expected Hamming accuracy stays within the estimated expected accuracies for all ranges except for (0.7, 0.75] for High Prob, (0.65, 0.7] for Shannon Entropy, (0.65, 0.7], (0.8, 0.85] for Collision Entropy, and (0.65, 0.7] for the mixture of each candidate function. Each of the four methods of predicting the expected accuracy have performed well. Collision Entropy shows to have the most interval matches (shown in bold), but it provides an overestimate in the range of (0.8, 0.85] which is not desired. Shannon Entropy provides the next most interval matches.

Next examining the Jaccard accuracy interval table (Table 7), the expected Jaccard accuracy lies within the computed accuracy bounds for (0.4, 0.5] and (0.2, 0.3] using HP, SE and CE and (0.9, 1] for the mixture of confidence functions. Each model does provide a greater true expected Jaccard accuracy for each computed confidence range, except for HP, SE and CE for the interval (0.9, 1] and all models

TABLE 6: The 95% interval for the expected Hamming accuracy for the given range of predicted Hamming confidence when modelled on each of High Prob (HP), Shannon Entropy (SE), Collision Entropy (CE) and a mixture of all confidence functions (MIX). The bold expected Hamming intervals are the closest match to the predicted intervals. The underlined intervals have a lower bound that is lower than the predicted interval.

Exp Acc	HP	SE	CE	MIX
(0.65,0.7]	NA	0.688, 0.746	0.657, 0.754	0.641, 0.737
(0.7,0.75]	0.720, 0.772	0.707, 0.742	0.720, 0.747	0.709, 0.733
(0.75,0.8]	0.750, 0.764	0.767, 0.782	0.765, 0.789	0.763, 0.780
(0.8,0.85]	0.808, 0.830	0.815, 0.848	0.792, 0.824	0.822, 0.837
(0.85,0.9]	0.873, 0.885	0.866, 0.884	0.864, 0.882	0.871, 0.888
(0.9,0.95]	0.932, 0.940	0.923, 0.938	0.925, 0.945	0.929, 0.935
(0.95,1]	0.959, 0.962	0.955, 0.959	0.957, 0.965	0.961, 0.964

TABLE 7: The 95% interval for the expected Jaccard accuracy for the given range of predicted Jaccard confidence when modelled on each of High Prob (HP), Shannon Entropy (SE), Collision Entropy (CE) and a mixture of all confidence functions (MIX). The bold expected Jaccard intervals are the closest match to the predicted intervals. The underlined intervals have a lower bound that is lower than the predicted interval.

Exp Acc	HP	SE	CE	MIX
(0,0.1]	0.362, 0.399	0.362, 0.398	0.362, 0.399	0.328, 0.374
(0.1,0.2]	0.185, 0.363	0.196, 0.301	0.183, 0.329	0.092, 0.136
(0.2,0.3]	0.236, 0.257	0.226, 0.263	0.246, 0.280	0.259, 0.308
(0.3,0.4]	0.384, 0.480	0.431, 0.483	0.383, 0.462	0.385, 0.429
(0.4,0.5]	0.450, 0.495	0.449, 0.488	0.439, 0.483	0.474, 0.526
(0.5,0.6]	0.579, 0.639	0.596, 0.679	0.584, 0.642	0.592, 0.647
(0.6,0.7]	0.677, 0.771	0.779, 0.861	0.726, 0.872	0.698, 0.772
(0.7,0.8]	0.880, 0.953	0.894, 0.956	0.909, 0.960	0.845, 0.895
(0.8,0.9]	0.904, 0.968	0.889, 0.949	0.858, 0.960	0.924, 0.973
(0.9,1]	<u>0.785, 1.000</u>	<u>0.817, 0.952</u>	<u>0.771, 1.000</u>	0.913, 0.968

for the interval (0, 0.1]. The mixture is preferred in this case, due to it providing a large number of interval matches, and not overestimating the high accuracy values.

Finally, the Exact Match interval table (Table 8) provides the expected Exact Match for each of the models' computed intervals. The correct predicted intervals were provided at (0.2, 0.3] for HP, (0.1, 0.2] for SE, (0.2, 0.3] and (0.1, 0.2] for CE, and (0.4, 0.5], (0.3, 0.4], (0.1, 0.2] and (0, 0.1] for the mixture. Each expected accuracy interval was within the range of the computed interval, or greater than the interval for all but three cases for HP, two cases for SE, three cases for CE and two cases for the mixture. For the two cases where the mixture overestimates, the error is small (for the predicted expected accuracy interval (0.9, 1.0] the expected Exact Match accuracy interval is (0.887, 0.940], and for the predicted expected accuracy of (0.7, 0.8] the expected accuracy interval is (0.691, 0.806]). Again, we prefer the Mixture, due to the most interval matches and the fewest overestimates.

Figure 3 presents the expected accuracy for a given predicted expected accuracy score. The ideal expected accuracy function will provide a straight line travelling along the diagonal, meaning that the estimated expected accuracy has a one to one mapping with the true expected accuracy.

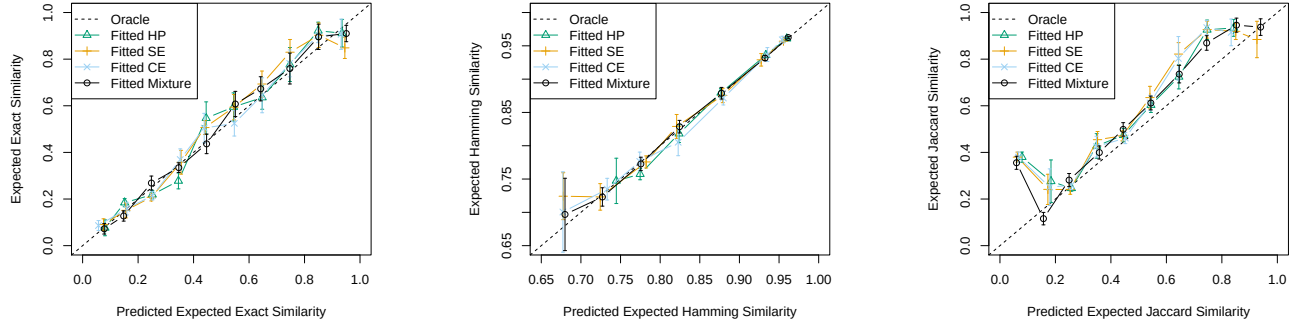


Fig. 3: The expected accuracy versus the predicted expected accuracy for the Exact, Hamming and Jaccard Similarity functions, using the candidate functions HP, SE, CE, or a mixture of all. The error bars show the 95% confidence interval for the expected accuracy. The Oracle line shows the ideal values.

TABLE 8: The 95% interval for the expected Exact Match accuracy for the given range of predicted Exact Match confidence when modelled on each of High Prob (HP), Shannon Entropy (SE), Collision Entropy (CE) and a mixture of all confidence functions (MIX). The bold expected Exact Match intervals are the closest match to the predicted intervals. The underlined intervals have a lower bound that is lower than the predicted interval.

Exp Acc	HP	SE	CE	MIX
(0.0,1]	0.051, 0.105	0.066, 0.108	0.066, 0.106	0.056, 0.094
(0.1,0.2]	0.171, 0.201	0.126, 0.174	0.134, 0.179	0.108, 0.145
(0.2,0.3]	0.205, 0.236	0.199, 0.222	0.203, 0.231	0.254, 0.305
(0.3,0.4]	0.256, 0.314	0.317, 0.401	0.334, 0.412	0.319, 0.353
(0.4,0.5]	0.491, 0.598	0.479, 0.528	0.451, 0.555	0.402, 0.474
(0.5,0.6]	0.555, 0.659	0.541, 0.638	0.490, 0.583	0.560, 0.648
(0.6,0.7]	0.579, 0.670	0.652, 0.750	0.568, 0.700	0.617, 0.710
(0.7,0.8]	0.724, 0.839	0.786, 0.880	0.728, 0.815	0.691, 0.806
(0.8,0.9]	0.893, 0.955	0.842, 0.931	0.873, 0.924	0.835, 0.924
(0.9,1]	0.860, 0.956	0.820, 0.894	0.854, 0.964	0.887, 0.940

TABLE 9: Summary of key findings in terms of accuracy, robustness and mixture for each expected accuracy candidate.

	Accuracy	Robustness	Mix
Exact	HP, SE, CE, ME, CS	CE, CS $\succ$ SE $\succ$ HP, ME	✓
Hamming	HP, SE, CE, ME, CS	CE, CS $\succ$ SE $\succ$ HP, ME	✗
Jaccard	SE	CE, CS $\succ$ SE $\succ$ HP, ME	✓

We see that the Mixture curve is always the closest to the diagonal line, or has small error, where the other functions may be close to the diagonal at some points, but have large deviations at other points. Both Exact and Hamming similarities have close fits to the diagonal, but Jaccard has a large error at the lower end. It is unclear why this has occurred, but being at the lower end means that it will not impact the use of the expected accuracy functions.

In summary, we have found that when fitting the expected accuracy function (consisting of the candidate functions) to data, any of the candidate functions will provide a good model for Hamming Similarity, and a mixture of the set of candidate functions provides the best estimates for Jaccard and Exact Match.

7 CONCLUSION

Probabilistic multi-label classification provides us with a distribution over the power set of labels. The probability of the mode labelset is commonly used as an estimate of the prediction confidence, however this measurement is not directly dependent on the chosen multi-label evaluation metric. Therefore, when taking into account the evaluation metric, the notion of confidence is not clear. Expected accuracy provides the practitioner with a measurement of the estimated accuracy with respect to the evaluation metric.

We hypothesised that expected accuracy of a multi-label prediction is a function of the categorical distribution over the powerset of the labels conditioned on the given observation. In this article we examined the correlation and robustness of seven candidate functions for measuring the expected accuracy of each prediction using the multi-label distribution. We found that the candidate functions Collision Entropy, Shannon Entropy and High Prob were highly correlated to expected accuracy (and the closely related Chi-squared and Min Entropy) and were also robust to changes in variables (such as changes in data) in the given order.

When fitting a model of expected accuracy with respect to different similarity (performance, evaluation) metrics, we found that High Prob, Shannon Entropy and Collision Entropy each provided high accuracy estimates for Hamming expected accuracy, and a mixture of these three functions provided high accuracy estimates for Jaccard and Exact expected accuracy. These findings (presented in Table 9) allow high accuracy estimates of the expected accuracy which can be in turn used to assess confidence in multi-label model results.

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APPENDIX

.1 Candidate accuracy function normalisation

We stated that each accuracy function must provide a score of 0 when the distribution is uniform and a score of 1 when one element has a probability of 1. Here we present how each function was normalised to provide this.

.1.1 High probability (HP)

High probability is given as:

$$H_{HP} = \max_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i) \quad (20)$$

To remove the dependence of this function on the label set size, we normalise the function so that the maximum value of 1 is given when all but one item has zero probability, and minimum value of 0 is given when all items have equal probability;

$$C_{HP} = \max_{\mathbf{y}_i \in \{0,1\}^L} \frac{2^L P(\mathbf{y}_i) - 1}{2^L - 1} \quad (21)$$

where L is the number of labels. For example, for 3 labels, a uniform distribution of $P(\mathbf{y}_i) = \frac{1}{8}$ (since there are 8 possible combinations) gives $C_{HP} = \frac{2^3 \cdot \frac{1}{8} - 1}{2^3 - 1} = 0$; the lowest expected accuracy.

.1.2 Top Gap (TG)

Top Gap is given as:

$$C_{TG} = \left[\max_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i) \right] - \left[\max_{\mathbf{y}_i \in \{\{0,1\}^L \setminus \mathbf{y}^*\}} P(\mathbf{y}_i) \right] \quad (22)$$

$$= P(\mathbf{y}^*) - \left[\max_{\mathbf{y}_i \in \{\{0,1\}^L \setminus \mathbf{y}^*\}} P(\mathbf{y}_i) \right] \quad (23)$$

This function is already appropriately limited between 0 and 1, therefore no normalisation is needed.

.1.3 Shannon Entropy (SE)

Shannon entropy is given as:

$$H_{SE} = - \sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i) \log(P(\mathbf{y}_i)) \quad (24)$$

To adjust the range to $[0, 1]$, we can use normalised entropy:

$$H_{SE}^* = - \sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i) \frac{\log(P(\mathbf{y}_i))}{\log(2^L)} \quad (25)$$

$$= - \sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i) \log_{2^L}(P(\mathbf{y}_i)) \quad (26)$$

where $H_{SE}^* = 1$ is provided when the probability of all class combinations are equal [27]. Note that normalised entropy is independent of the entropy log base, since the normalisation converts the base to 2^L .

To obtain an expected accuracy candidate, we subtract the entropy score from 1 to obtain

$$C_{SE} = 1 + \sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i) \log_{2^L}(P(\mathbf{y}_i)) \quad (27)$$

where $C_{SE} \in [0, 1]$ is a measure of expected accuracy using Shannon Entropy ($C_{SE} = 0$ implies no confidence and low expected accuracy, while $C_{SE} = 1$ implies full confidence and high expected accuracy for the prediction).

.1.4 Collision Entropy (CE)

Collision entropy (CE) is given as:

$$H_{CE} = - \log \sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i)^2 \quad (28)$$

Collision entropy can be normalised to provide an expected accuracy candidate, giving a score of zero when the distribution is Uniform, and a score of 1 when all but one of the label combinations are zero:

$$C_{CE} = 1 + \log_{2^L} \sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i)^2 \quad (29)$$

.1.5 Min Entropy (ME)

Min Entropy (ME) is given as:

$$H_{ME} = - \log \max_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i) \quad (30)$$

This entropy measurement can be normalised to obtain a potential measure of expected accuracy:

$$C_{ME} = 1 + \log_{2^L} \max_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i) \quad (31)$$

where $C_{ME} \in [0, 1]$, $C_{ME} = 0$ implies low expected accuracy, while $C_{ME} = 1$ implies high expected accuracy (full confidence in the prediction).

.1.6 Gini Impurity

Gini impurity (GI) is given as:

$$H_{GI} = \sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i)(1 - P(\mathbf{y}_i)) \quad (32)$$

A normalised form of Gini impurity provides us with an potential measure of expected accuracy:

$$C_{GI} = 1 - \sum_{\mathbf{y}_i \in \{0,1\}^L} \frac{P(\mathbf{y}_i)(1 - P(\mathbf{y}_i))}{1 - 2^{-L}} \quad (33)$$

where $C_{GI} = 0$ when all labels have the same probability, and $C_{GI} = 1$ when all but one item have zero probability.

.1.7 Chi-squared Statistic

For our use of the Chi-squared Statistic, we compare probabilities rather than frequencies, and we set the expected distribution to be Uniform, so the further the computed distribution is from Uniform, the greater the score.

The Chi-squared statistic (CS) is given as:

$$H_{CS} = \sum_{\mathbf{y}_i \in \{0,1\}^L} \frac{(P(\mathbf{y}_i) - 2^{-L})^2}{2^{-L}} \quad (34)$$

The normalised Chi-square statistic ensures a maximum of 1 when all but one of the items have zero probability, and a score of 0 when all items have equal probability.

$$C_{CS} = \sum_{\mathbf{y}_i \in \{0,1\}^L} \frac{(P(\mathbf{y}_i) - 2^{-L})^2}{1 - 2^{-L}} \quad (35)$$

TABLE 10: Correlation between the chosen acceptance function and the Exact Match accuracy for each data set and each multi-label method. Arrows show a significant difference at the 0.1 (one arrow), 0.05 (two arrows) and 0.01 (three arrows) levels, when compared to HP.

Method	HP, ME	TG	SE	CE, CS, GI
<i>Emotion</i>				
ECC	0.2595	0.1695↓↓↓	0.2345	0.2523
CT	0.2658	0.1442↓↓	0.2133↓↓	0.2362↓
Indep	0.1272	0.0409↓	0.1636	0.1568
<i>Scene</i>				
ECC	0.3440	0.3365	0.3033↓↓↓	0.3276↓↓↓
CT	0.3073	0.2676↓↓↓	0.3424↑↑↑	0.3222↑↑
Indep	0.2690	0.2106↓↓↓	0.3407↑↑↑	0.2994↑↑↑
<i>Yeast</i>				
ECC	0.3167	0.3066	0.2789↓↓↓	0.2949↓↓↓
CT	0.3117	0.1978↓↓↓	0.3126	0.3171
Indep	0.3217	0.2448↓↓↓	0.2548↓↓↓	0.3383↑↑
<i>Stare</i>				
ECC	0.2651	0.2506	0.1813↓↓	0.2384
CT	0.2452	0.2079	0.2821	0.2490
Indep	0.0889	0.0526↓↓↓	0.0927	0.0939↑
<i>Enron</i>				
ECC	0.3223	0.2983↓↓	0.2694↓↓	0.3242
CT	0.3122	0.2904↓	0.2662↓	0.3049
Indep	0.3191	0.2981↓↓	0.3483	0.3504↑↑
<i>Slashdot</i>				
ECC	0.1886	0.1018↓↓↓	0.2618↑↑↑	0.2472↑↑↑
CT	0.1030	0.0219↓↓↓	0.2316↑↑↑	0.1652↑↑↑
Indep	0.0546	-0.0208↓↓↓	0.1993↑↑↑	0.1196↑↑↑

.2 Situational Correlation

Relative acceptance implies that the order provided by accuracy and acceptance should be the same; i.e., highly correlated. To measure this correlation (i.e., ordinal similarity), we use Kendall's τ correlation. Results are presented in Tables 10, 11, and 12 for the different metrics. Each table presents the correlation for the given similarity function broken down by data set and method.

The results show that Top Gap (TG) is either equivalent to, or worse than High Prob (HP) in all cases. Also we note that Shannon Entropy and the Collision Entropy equivalents are superior under Slashdot. For all other combinations, the results are mixed; there are cases when HP has the highest correlation and cases where the other candidate acceptance function have higher correlation.

We conclude that candidate acceptance functions other than HP are more appropriate under given conditions. This is reinforced by Figure 4, which provides a graphical representation of the CT method results from Table 12.

.3 Optimising with respect to a similarity function

The multi-label problem requires us to find

$$\hat{\mathbf{y}} = \arg \max_{\mathbf{y}_i \in \{0,1\}^L} E_S(\mathbf{y}_i) \quad (36)$$

TABLE 11: Correlation between the chosen acceptance function and Hamming accuracy for each data set and each multi-label method. Arrows show a significant difference at the 0.1 (one arrow), 0.05 (two arrows) and 0.01 (three arrows) levels, when compared to HP.

Method	HP, ME	TG	SE	CE, CS, GI
<i>Emotion</i>				
ECC	0.2527	0.1549↓↓↓	0.2280	0.2434
CT	0.2245	0.0506↓↓↓	0.2013	0.2212
Indep	0.1223	0.0216↓↓↓	0.1593↑	0.1503
<i>Scene</i>				
ECC	0.346	0.331↓	0.302↓↓↓	0.330↓↓↓
CT	0.302	0.260↓↓↓	0.328↑↑	0.315↑↑
Indep	0.264	0.205↓↓↓	0.320↑↑↑	0.290↑↑↑
<i>Yeast</i>				
ECC	0.2948	0.2610↓↓	0.2733↓	0.2859
CT	0.2492	0.1176↓↓↓	0.2634	0.2668↑↑
Indep	0.2399	0.1528↓↓↓	0.1534↓↓↓	0.2719↑↑↑
<i>Stare</i>				
ECC	0.2818	0.2497	0.2301	0.2738
CT	0.2040	0.1537↓↓	0.2469	0.2195
Indep	0.1568	0.0958↓↓	0.1738	0.1662
<i>Enron</i>				
ECC	0.3204	0.2644↓↓↓	0.3200	0.3568↑↑
CT	0.3023	0.2513↓↓↓	0.2988	0.3261
Indep	0.3188	0.2640↓↓↓	0.3951↑↑↑	0.3842↑↑↑
<i>Slashdot</i>				
ECC	0.2003	0.1236↓↓↓	0.2537↑↑↑	0.2493↑↑↑
CT	0.1220	0.0493↓↓↓	0.2256↑↑↑	0.1762↑↑↑
Indep	0.0778	0.0107↓↓↓	0.1927↑↑↑	0.1336↑↑↑

TABLE 12: Correlation between the chosen acceptance function and Jaccard accuracy for each data set and each multi-label method. Arrows show a significant difference at the 0.1 (one arrow), 0.05 (two arrows) and 0.01 (three arrows) levels, when compared to HP.

Method	HP, ME	TG	SE	CE, CS, GI
<i>Emotion</i>				
ECC	0.3046	0.1692↓↓↓	0.2700↓	0.2945
CT	0.2989	0.0694↓↓↓	0.2491↓↓	0.2857
Indep	0.1927	0.0709↓↓↓	0.2115	0.2138
<i>Scene</i>				
ECC	0.348	0.335↓	0.311↓↓↓	0.333↓↓↓
CT	0.310	0.259↓↓↓	0.356↑↑↑	0.329↑↑↑
Indep	0.263	0.194↓↓↓	0.349↑↑↑	0.298↑↑↑
<i>Yeast</i>				
ECC	0.2865	0.2521↓↓↓	0.2634↓	0.2761
CT	0.2762	0.1295↓↓↓	0.2914	0.2976↑↑
Indep	0.2586	0.1449↓↓↓	0.1306↓↓↓	0.3051↑↑↑
<i>Stare</i>				
ECC	0.2941	0.2828	0.1992↓↓	0.2633
CT	0.2139	0.1778	0.2993↑↑↑	0.2413
Indep	0.1377	0.0879	0.2073↑↑	0.1620
<i>Enron</i>				
ECC	-0.1284	-0.1160	0.0648↑↑↑	-0.0509↑↑↑
CT	-0.1238	-0.1125	0.1254↑↑↑	-0.0333↑↑↑
Indep	-0.1509	-0.1231	-0.0628↑↑↑	-0.1115
<i>Slashdot</i>				
ECC	0.1802	0.0892↓↓↓	0.2671↑↑↑	0.2442↑↑↑
CT	0.0849	0.0010↓↓↓	0.2339↑↑↑	0.1541↑↑↑
Indep	0.0329	-0.0445↓↓↓	0.1954↑↑↑	0.1040↑↑↑

By maximising with respect to Exact Match, we obtain the mode of the labelset distribution.

$$\begin{aligned} \hat{\mathbf{y}} &= \arg \max_{\mathbf{y}_i \in \{0,1\}^L} \mathbb{E}_{\mathbf{Y} \sim P(\mathbf{Y}|\mathbf{x})} [\text{EM}(\mathbf{y}_i, \mathbf{Y})|\mathbf{x}] \\ &= \arg \max_{\mathbf{y}_i \in \{0,1\}^L} \left(\sum_{\mathbf{y}_j \in \{0,1\}^L} \text{EM}(\mathbf{y}_i, \mathbf{y}_j) P(\mathbf{y}_j|\mathbf{x}) \right) \\ &= \arg \max_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i|\mathbf{x}) = \mathbf{y}^* \end{aligned}$$

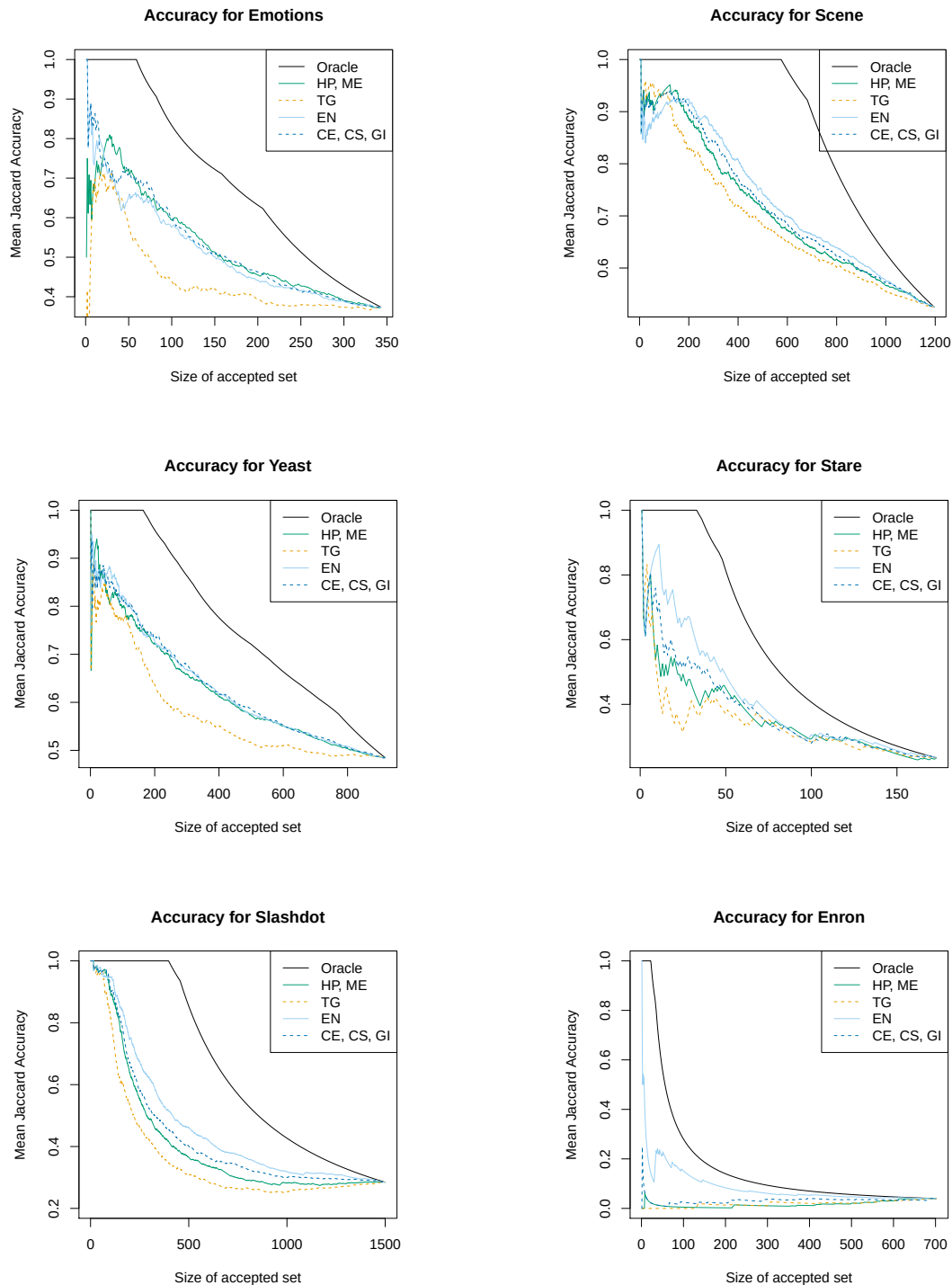


Fig. 4: Mean Jaccard accuracy when choosing only the predictions with greatest acceptance score when using a Classifier Trellis over the six data sets. The x -axis shows the number of predictions accepted (e.g. 1 implies that only the prediction with the greatest acceptance score was included in the mean, and the right most value is when including all predictions in the mean accuracy).

therefore, equation (36) is equivalent to equation (1) when using Exact Match. This is also seen in Table 2.

If we were able to assume independent labels we can expand $P(\mathbf{y}_i|\mathbf{x})$ from equation (1) as a product of independent elements and move the arg max inside, to obtain

$$\begin{aligned}\hat{\mathbf{y}} &= \arg \max_{y_1, \dots, y_L} [P(y_1|\mathbf{x}) \cdots P(y_L|\mathbf{x})] \\ &= [\arg \max_{y_1 \in \{0,1\}} P(y_1|\mathbf{x}), \dots, \arg \max_{y_L \in \{0,1\}} P(y_L|\mathbf{x})] \quad (37)\end{aligned}$$

If using Hamming Similarity, we obtain,

$$\begin{aligned}\hat{\mathbf{y}} &= \arg \max_{\mathbf{y}_i \in \{0,1\}^L} \mathbb{E}_{\mathbf{Y} \sim P(\mathbf{Y}|\mathbf{x})} [\text{HS}(\mathbf{y}_i, \mathbf{y})] \\ &= \arg \max_{\mathbf{y}_i \in \{0,1\}^L} \left(\sum_{\mathbf{y}_j \in \{0,1\}^L} \text{HS}(\mathbf{y}_i, \mathbf{y}_j) P(\mathbf{y}_j|\mathbf{x}) \right) \\ &= \arg \max_{\mathbf{y}_i \in \{0,1\}^L} \left(\sum_{\mathbf{y}_j \in \{0,1\}^L} \frac{1}{L} \sum_{l=1}^L [y_{il} = y_{jl}] P(\mathbf{y}_j|\mathbf{x}) \right) \\ &= \arg \max_{\mathbf{y}_i \in \{0,1\}^L} \left(\sum_{l=1}^L \sum_{\mathbf{y}_j \in \{0,1\}^L} [y_{il} = y_{jl}] P(\mathbf{y}_j|\mathbf{x}) \right) \\ &= \arg \max_{\mathbf{y}_i \in \{0,1\}^L} \left(\sum_{l=1}^L \mathbb{E}_{\mathbf{Y} \sim P(\mathbf{Y}|\mathbf{x})} [y_{il} = y_l] \right)\end{aligned}$$

The expected value of each label being 0 or 1, is positive, so we can compute the argmax of each element of the sum to obtain the prediction for each label:

$$\hat{y}_l = \arg \max_{z \in \{0,1\}} (\mathbb{E}_{\mathbf{Y} \sim P(\mathbf{Y}|\mathbf{x})} [z = y_l]) \quad \text{for each label } l \quad (38)$$

which relates to Eq. (37); showing that the chosen labelset consists of the mode for each label (marginal mode). It is worth remarking that the sum comes outside the expectation in the derivation of expected Hamming similarity.

.4 Equivalence of candidate expected accuracy functions when using relative expected accuracy.

Gini impurity is given as:

$$\begin{aligned}C_{\text{GI}} &= 1 - \sum_{\mathbf{y}_i \in \{0,1\}^L} \frac{P(\mathbf{y}_i)(1 - P(\mathbf{y}_i))}{1 - 2^{-L}} \\ &= 1 - \sum_{\mathbf{y}_i \in \{0,1\}^L} \frac{P(\mathbf{y}_i) - P(\mathbf{y}_i)^2}{1 - 2^{-L}} \\ &= 1 - \frac{\sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i)}{1 - 2^{-L}} + \frac{\sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i)^2}{1 - 2^{-L}} \\ &= 1 - \frac{1}{1 - 2^{-L}} + \frac{\sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i)^2}{1 - 2^{-L}} \\ &= \frac{-2^{-L}}{1 - 2^{-L}} + \frac{\sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i)^2}{1 - 2^{-L}}\end{aligned}$$

which provides an equivalent ordering to $\sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i)^2$. By examining equation 29, and again noting that $1 + \log_{2L}(x)$ is monotonic, we find that CE has

an equivalent ordering to $\sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i)^2$ and hence also to Gini impurity. Also

$$\begin{aligned}C_{\text{CS}} &= \sum_{\mathbf{y}_i \in \{0,1\}^L} \frac{(P(\mathbf{y}_i) - 2^{-L})^2}{1 - 2^{-L}} \\ &= \sum_{\mathbf{y}_i \in \{0,1\}^L} \frac{P(\mathbf{y}_i)^2 - 2^{1-L}P(\mathbf{y}_i) + 2^{-2L}}{1 - 2^{-L}} \\ &= -\frac{2^{1-L}}{1 - 2^{-L}} + \frac{2^{-L}}{1 - 2^{-L}} + \sum_{\mathbf{y}_i \in \{0,1\}^L} \frac{P(\mathbf{y}_i)^2}{1 - 2^{-L}} \\ &= -\frac{2^{-L}}{1 - 2^{-L}} + \sum_{\mathbf{y}_i \in \{0,1\}^L} \frac{P(\mathbf{y}_i)^2}{1 - 2^{-L}}\end{aligned}$$

provides an equivalent ordering to $\sum_{\mathbf{y}_i \in \{0,1\}^L} P(\mathbf{y}_i)^2$ and is identical to C_{GI} . Showing that C_{GI} , C_{CE} and C_{CS} provide the same ordering.

.5 Relative and Absolute Association

The candidate expected-accuracy functions described above (Section 4.1; generically stated in equation (9)) provide a measure f for the labelset distribution of each instance. This score provided by f has the potential to be used in practice to determine the suitability of the model for the given instance $\mathbf{x}$ (i.e., to estimate the expected accuracy of its prediction for that instance), with higher values (closer to 1) implying that the model is more suitable. For the score to be useful it must have some association to the actual accuracy of the prediction. Therefore, the candidate functions will be assessed based on their association to the true expected accuracy.

We discuss two levels of expected accuracy association: a weaker form of association that we call *relative association* and stronger form that we call *absolute association*.

Let $\mathcal{D} = \{(\mathbf{x}_n, \hat{\mathbf{y}}_n)\}_{n=1}^N$ be a data set of N test instances alongside their predictions. Let $r_n \in \{1, \dots, N\}$ indicate the ranking of the n -th instance, such that $r(f) := r_1, \dots, r_N$ is a permutation of $1, \dots, N$, corresponding to rank *when scored by* candidate function f . In this context, $r_1(f)$ is the index of the instance providing highest score $f(P(\mathbf{y}|\mathbf{x}_{r_1}))$, and so on. Likewise, let $r_1(E)$ correspond to the index of the instance giving highest expected accuracy (as per equation (8)), i.e., $E(\mathbf{y}_{r_1})$ is the highest value, and $E(\mathbf{y}_{r_N})$ is the lowest. Recall that the expected accuracy is with respect to a similarity function of interest, $\mathcal{S}$.

Relative Association: A candidate function f provides a *relative association* to expected accuracy (with respect to a given similarity function) if the ordering of the instances $r_1(f), \dots, r_N(f)$ provides a close match (e.g., monotonically increasing) to the ordering of the instances provided by their expected accuracy $r_1(E), \dots, r_N(E)$. In other words we measure

$$\text{RelativeAssociation}(r(f), r(E))$$

where f and E are both associated with some similarity function $\mathcal{S}$. The association will not necessarily be the same for different $\mathcal{S}$. An example is given in Table 13.

TABLE 13: An example for $N = 3$ instances, $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$, supposing $f := f_{\text{ME}}$ (i.e., using Min Entropy) and $E := E_{\text{HS}}$ (i.e., using Hamming Similarity). In relative association we compare the ranks (columns $r(E)$ vs $r(f)$). In absolute association, we compare columns E and f . In experiments we do this for different classes of f and E ; and different similarity metrics $\mathcal{S}$.

i	$\mathbf{x}_i$	$\mathbf{y}_i = [Y_1 \ Y_2 \ Y_3]$	E	f	$r(E)$	$r(f)$
1	$\mathbf{x}_1$	[0 0 0]	0.6	0.2	1	3
2	$\mathbf{x}_2$	[0 0 1]	0.5	0.8	2	2
3	$\mathbf{x}_3$	[0 1 0]	0.5	3.1	2	1

Absolute Association: A candidate function f provides an *absolute association* to expected accuracy if the scores provided by each are associated. In other words we measure

$$\text{AbsoluteAssociation}(f, E)$$

Absolute association is stronger, since it implies that we are able to predict the expected accuracy from the candidate function. Whereas relative association only provides us with which instances have a greater expected accuracy than others (suggesting that there is a monotonic relationship).

Analysis of Relative association allows us to determine if an association exists. This initial investigation is found in Appendix F.

When measuring relative expected accuracy, we examine the order (ranking) of the scores, not the values of the scores, therefore some of the candidate expected accuracy functions become equivalent. We find that both CE and Gini impurity, and Min entropy and High probability provide equivalent orderings/rankings

With regard to min entropy (given in equation (14)), the terms $1 + y$ and $\log_{2L}(y)$ are monotonic functions which do not effect the order of y . Therefore the ordering provided by C_{ME} is equivalent to the ordering provided by C_{HP} . Further details are provided in Appendix D.

Therefore, when analysing relative accuracy, we examine only HP, TG, SE and CE. When analysing absolute accuracy, we examine HP, TG, SE, CE, ME and CS.

.6 Relative Expected Accuracy

A candidate function provides high relative expected accuracy if it is weakly monotonically increasing with respect to the chosen evaluation score, implying an association. To examine if an expected accuracy candidate provided a weak increasing monotonic relationship to the evaluation scores, the Kendall's τ correlation between the accuracy and candidate scores was computed for each combination of variables: candidate function, multi-label method, data set and evaluation function, giving 216 correlation results. These results are presented in Appendix B using HP as the baseline. The results show that TG is generally the same or has lower correlation than HP, but the for remaining candidate functions there are some cases that have higher correlation, some the same and some worse.

According to these results, it is clear that the data, method and metric variables effect the correlation, but it is not clear *how* they effect the correlation. To obtain insight

into the effect of each variable on the [accuracy – candidate score] correlation, we provide a marginal analysis for accuracy and robustness.

.6.1 Analysis 4: Accuracy Analysis for Relative Accuracy

Having found that there is merit in the candidate expected accuracy functions other than HP in particular conditions, we now delve deeper and generalise the results. In particular, we examine the marginal effect of each experimental variable (candidate function, dataset and method – under each metric) on the correlation.

We use Fisher's Z transformation to provide variance stabilisation for the correlation values to allow us to model the effect of each of the variables on the correlation using a linear fixed effects model. The results are presented in Table 14.

Notice the high R^2 value and small RMSE values for each model, showing that the deviation of the data to this analytical model is small. This implies that the analytical model is appropriate for the correlation data analysis.

The table shows each variable (Expected Accuracy Function, Data, Classifier) using the baseline values High Prob for Function, Emotions for Data and ECC for Classifier. The entries in the table show the bias (change in mean variance stabilised Kendall's tau correlation) when changing the variable from the baseline to the given value. A positive value implies that the correlation has increased, implying that the candidate function and accuracy are closer to being weakly monotonic, while a negative value implies a reduction in correlation.

The table shows that TG reduces the correlation (shown by its negative coefficient), and that Shannon Entropy and Collision Entropy increase the correlation (shown by their positive coefficients) for each of the Exact, Hamming and Jaccard similarities. The stars associated to each coefficient show the range of the p-value for the hypothesis test, testing if there is evidence that the coefficient is not zero. No significance implies that there is no evidence and hence using the acceptance method is no different to the baseline, and significance implies a difference to the baseline HP.

The results imply that we should be using HP for Exact, Shannon Entropy for Jaccard and Collision Entropy for Hamming similarity.

The Data section shows that changing the data set does have an effect on the expected correlation, with Enron having the greatest effect.

The Classifier section indicates that using an Independent classifier leads to significantly worse correlation at the 1% level for Exact, Hamming and Jaccard. CT also provides a drop in expected correlation, but is only significant at the 1% level when using Hamming similarity. Since we generally expect that the order from lowest to highest (due to the complexity of each method) would be Independent, CT, ECC, the analysis shows that a more accurate classifier leads to higher expected correlation.

Note that since the analytical model is modelling each variable independently, the significant results for Data and Classifier do not impact the significant results for Expected Accuracy functions.

TABLE 14: The expected change in variance stabilised Kendall’s τ correlation for the three variables (Expected Accuracy function, Data, Classifier) and standard error in parentheses, with respect to each variable’s baseline, independent of the other variable values. A value of 0 represents no change in the correlation when changing from the baseline to the chosen item, a positive value shows an increase in correlation, a negative value implies a decrease in correlation. The stars represent the statistical significance level of each expected change. The footer of the table shows the goodness of fit statistics for the analytical model.

	Exact	Jaccard	Hamming
<i>Expected Accuracy Function Bias (Baseline: High Prob)</i>			
Top Gap	−0.058 (0.017)***	−0.070 (0.018)***	−0.073 (0.013)***
Shannon Entropy	0.008 (0.017)	0.050 (0.018)***	0.017 (0.013)
Collision Entropy	0.012 (0.017)	0.030 (0.018)*	0.023 (0.013)*
<i>Data Bias (Baseline: Emotions)</i>			
Scene	0.124 (0.021)***	0.094 (0.022)***	0.138 (0.016)***
Yeast	0.109 (0.021)***	0.024 (0.022)	0.070 (0.016)***
Stare	−0.001 (0.021)	−0.006 (0.022)	0.036 (0.016)**
Enron	0.128 (0.021)***	−0.293 (0.022)***	0.157 (0.016)***
Slashdot	−0.050 (0.021)**	−0.094 (0.022)***	−0.018 (0.016)
<i>Classifier Bias (Baseline: ECC)</i>			
Independent	−0.069 (0.015)***	−0.085 (0.015)***	−0.074 (0.011)***
CT	−0.020 (0.015)	−0.027 (0.015)*	−0.046 (0.011)***
R ²	0.751	0.886	0.838
RMSE	0.051	0.053	0.039

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

6.2 Analysis 5: Robustness Analysis for Relative Accuracy

We now investigate the robustness of each candidate function when used for relative accuracy. The results of the analysis are in Table 15.

Interpretation is similar to Table 5: split into five sections, the first four containing the marginal effects and the last section containing goodness of fit statistics. Each estimated coefficient is accompanied with a standard error directly below it in parentheses.

We find that the least robust is TG, having statistically significant non-zero coefficients for all but the Hamming coefficient (meaning that if we change the data, classifier, or similarity function, the acceptance score will correlate differently to the prediction accuracy). The most robust is SE, having no statistically significant non-zero coefficients, followed closely by CE, having coefficients with significance only at the 10% level.

It is interesting to see that HP is affected by changing the classifier from ECC to Independent, or changing the accuracy metric to Jaccard. Also the sample gradients for Label Count, Label Card and Feature count are negative (as these increase in the data, the correlation will reduce), while the sample gradient for Label comb is positive (as the number of label combinations increase, the correlation will increase) for all acceptance parameters. However, note that many of these results are not statistically significant.

The goodness of fit statistics (shown at the bottom of the table) are all poor, especially for SE and CE. This is a good sign, showing that the correlation has little association with the data parameters, classifier and accuracy metric. We conclude this section noting that SE is the most robust in terms of Kendall’s correlation to accuracy.

TABLE 15: The expected change in variance stabilised Kendall’s τ correlation when adjusting four data features (Label Count, Label Comb, Label Card, Feature Count), or changing Classifier or Similarity Function (standard error in parentheses), independent of the other variable values. A value of 0 represents no change in the correlation when changing the Data Feature, or when changing from the baseline to the chosen item, a positive value shows in increase in correlation, a negative value implies a decrease in correlation. The stars represent the statistical significance level of each expected change. The footer of the table shows the goodness of fit statistics for the analytical model.

	HP	TG	SE	CE
<i>Data Features Expected Gradient (Zero gradient means no effect)</i>				
Label Count	−0.046 (0.023)*	−0.078 (0.022)***	−0.024 (0.019)	−0.038 (0.022)*
Label Comb	0.011 (0.007)*	0.022 (0.006)***	0.005 (0.005)	0.009 (0.006)
Label Card	−0.471 (0.292)	−0.959 (0.282)***	−0.208 (0.237)	−0.374 (0.277)
Feature Count	−0.001 (0.001)*	−0.002 (0.001)***	−0.000 (0.000)	−0.001 (0.001)
<i>Classifier Bias (Baseline: ECC)</i>				
Independent	−0.092 (0.034)***	−0.112 (0.032)***	−0.039 (0.027)	−0.061 (0.032)*
CT	−0.040 (0.034)	−0.077 (0.032)**	0.015 (0.027)	−0.023 (0.032)
<i>Similarity Function Bias (Baseline: Exact)</i>				
Hamming	−0.009 (0.034)	−0.023 (0.032)	−0.000 (0.027)	0.001 (0.032)
Jaccard	−0.077 (0.034)**	−0.088 (0.032)***	−0.035 (0.027)	−0.059 (0.032)*
R ²	0.436	0.478	0.224	0.311
RMSE	0.101	0.097	0.082	0.096

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$